EMIKA SANDINA

Computer Science Undergraduate

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ABOUT ME

CS undergraduate at the University of Westminster with hands-on experience building and shipping full-stack web applications and ML-powered solutions. Proven ability to lead projects from concept to deployment, working across React.js, Node.js, Express.js, Java and Python. Strong foundation in designing RESTful APIs, managing databases, and developing predictive models. Passionate about applying AI to solve real-world problems. Looking to contribute to a product engineering team where I can work on real-world projects and grow fast.

EDUCATION

BSc (Hons) Computer Science

University Of Westminster

2024 - Present

Foundation Certificate in Higher Education

Informatics Institute Of Technology

2023 - 2024

Results : **Distinction (Top Grade)**

G.C.E Ordinary Level Examination

Vidura College

2016 - 2022

Results : 8A 1C

PROJECTS

Axiom AI — AI-Powered Coding Mentor Platform (Full Stack Individual Project)

React.js, Express.js, OpenRouter API (DeepSeek), Monaco Editor, Node.js

- Built a full-stack AI coding mentor platform where users improve their problem-solving skills through structured, feedback-driven practice
- Designed a workflow that requires users to explain their approach before accessing the code editor, encouraging independent thinking rather than copying solutions
- Connected the OpenRouter API to evaluate user explanations and return step-by-step feedback.

DengueGuard (Full Stack + Machine Learning Group Project)

React.js, Express.js, Supabase, Python, XGBoost

- Led a team of 5 to design and build a full-stack web platform that allows users to report dengue cases and predict future weekly case counts.
- Designed UI/UX prototypes in Figma and built frontend pages and components using React.js, translating designs into responsive and user-friendly interfaces.
- Built RESTful APIs using Express.js and integrated Supabase for real-time database storage, connecting the frontend to the reporting and prediction features.
- Trained an XGBoost machine learning model on historical dengue data to forecast weekly case numbers.

Loan Approval & Eligibility Prediction System (Machine Learning Individual Project)

Python, Pandas, Scikit Learn, Ensemble Learning (KNN + LR), Decision Tree

- Performed software analysis on a dataset of 58,000+ bank clients, building machine learning models to predict whether a loan should be approved and how much a client is eligible for.
- Trained and compared three classification algorithms (KNN, Logistic Regression, Naïve Bayes), selecting KNN as the best performer based on precision and overall detection ability.
- Combined KNN and Logistic Regression into a soft-voting ensemble model, which outperformed all individual models in accurately identifying high-risk loan applications.
- Fine-tuned the KNN model using GridSearchCV to improve prediction reliability with minimal trade-off.
- Built and compared a fully-grown vs pruned Decision Tree for predicting loan amounts, with the pruned model generalizing significantly better on unseen client data.

TECHNICAL SKILLS

- Web Development: Next.js (Learning), React.js, Node.js, Express.js, HTML, CSS
- Programming Languages: Python, JavaScript, Java, C#
- Databases: MySQL, Supabase (PostgreSQL)
- Machine Learning & AI: Scikit-learn, Pandas, Classification, Regression, LLM APIs, Prompt Engineering
- Data Analysis & Modeling: Feature engineering, model evaluation, hyperparameter tuning, GridSearchCV
- Tools & Platforms: GitHub, Postman, Vercel, Netlify, Trello, Microsoft 365
- Creative Design and Development: Figma, Canva, CapCut, Filmora
- Concepts: Object-Oriented Programming, Agile Methodologies, API Design, Database Design
- Interested In - Rust, Spring Boot, Cloud Platforms, CI/CD, Docker, Kubernetes

SOFT SKILLS

- Team Management
- Communication
- Task Management
- Time Management
- Leadership
- Problem Solving

ACHEIVEMENTS

Hult Prize 2026 - Participant

- Presented the ML-powered dengue healthcare platform (Dengue Guard).
- Demonstrated real-world application of predictive modeling in public health.

REFERENCES

Subhaka Bhavanishankar

Assistant Lecturer,

IIT

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Nuwan Weerathunga Arachchi

Associate Technical Lead,

Cambio Software

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